

Why an AI Summary Can Be Factually Accurate and Still Misleading

The Difference Between True Fragments and a Faithful Account

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Series Context

This paper applies the definitions in *What Is Semantic Fidelity*, *When Accuracy Is Not Enough*, and *The Compression Paradox in AI*. Semantic fidelity concerns the preservation of meaning across transformation. Semantic drift is gradual change in that meaning. Reality Drift is a broader system condition in which representations continue organizing action while losing corrective contact with what they represent.

Shared Illustrative Case

The following report is fictional and isolates the mechanism. A source report says:

“In this small observational sample, Clinic A had fewer repeat visits after introducing remote follow-up. Patients also received additional staffing support. The report cannot determine which change produced the difference, and the result should not be generalized to other clinics without further study.”

A later summary says:

“Remote follow-up reduced repeat visits at Clinic A.”

This paper examines one point at which the later sentence can be produced, accepted, or reused.

Core Claim

A summary can contain only facts found in its source and still change the conclusion a reasonable reader takes from that source.

Suppose an AI assistant is asked for a one-sentence account of the clinic report. It writes: *“Clinic A introduced remote follow-up and recorded fewer repeat visits.”* Both events appear in the report. The sentence does not explicitly claim that one caused the other. Yet in an answer to *“Does remote follow-up reduce repeat visits?”* it strongly suggests an affirmative answer. The prompt, selection, and placement do interpretive work even when every retained clause is true.

Now compare a second version: *“Remote follow-up reduced repeat visits at Clinic A.”* This version goes further. It substitutes a causal claim for an observation. The staffing change and the study’s uncertainty have disappeared. The difference between the two versions matters. The first can

mislead through framing and omission. The second also overstates what the evidence establishes. A useful analysis should identify the precise failure instead of calling both summaries simply inaccurate.

Meaning lives partly in relations among claims. The report says when the change occurred, what else changed, which result was observed, and what cannot be concluded. A summary must select material, but selection changes emphasis. If the conclusion and qualification are separated, the conclusion may travel without the qualification. A later reader sees a neat relationship between intervention and result where the source showed an unresolved question.

Factual accuracy tests the claims a summary makes. Source support tests whether particular claims can be traced to the report. Semantic fidelity asks whether the summary preserves the report's governing meaning for the intended use. These tests overlap, but success on one does not settle the others. A claim about the number of repeat visits might be numerically correct and still be used to support an inference the source withheld.

A practical comparison begins with four questions. What conclusion does the source license? How certain is it? To whom and under what conditions does it apply? What practical decision would a reasonable reader make after reading the summary alone? If the answer to the last question changes when the reader sees the source, the summary has lost something consequential. This comparison does not demand that every limitation appear in every summary. It asks that the limitations controlling the likely inference survive.

In the clinic case, a faithful short summary might read: *“Repeat visits fell at one clinic after remote follow-up and additional staffing were introduced; the report could not determine the cause.”* It leaves out many details while preserving the uncertainty that governs use of the result. A longer summary that repeatedly foregrounds remote follow-up and buries the staffing change could be less faithful despite containing more words.

The misleading summary becomes more consequential when it circulates. A reader may cite it as the source's conclusion. Another system may retrieve the citation and repeat the claim. The report remains somewhere in the archive, but the circulating representation increasingly determines what later readers believe the report said. Correction requires a reader or process capable of returning to the source and allowing its limits to alter the conclusion.

Diagnostic Question

Would a reasonable reader take the same conclusion, confidence, scope, and practical implication from the summary as from the source?

Connection to Reality Drift

One misleading summary changes meaning locally. Repeated use of that summary as evidence, especially when later readers cannot or do not return to the source, can contribute to Reality Drift. The representation remains coherent and traceable within the system while the conclusions it supports become progressively less constrained by the underlying evidence.